

School of Computer Science & IT

Department of BCA

SOFTWARE ENGINEERING (23BCA3C01)

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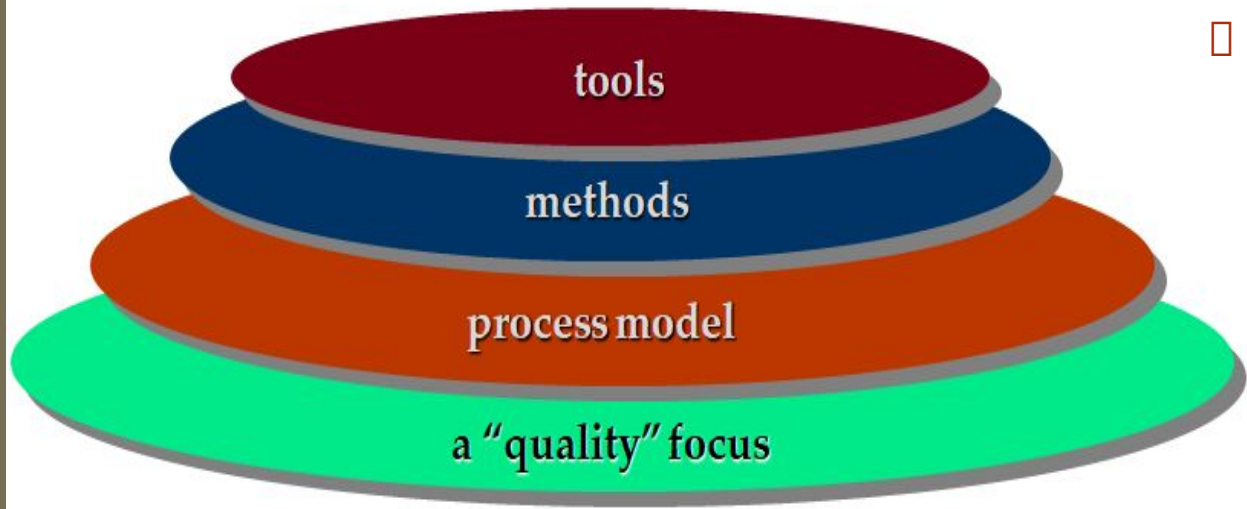
MODULE 1: Software Product and Process

- **Defining Software Engineering**
- **Software Process Framework**

Defining Software Engineering

- ❑ Software engineering is the systematic application of engineering approaches to the development of software.
- ❑ Before the product is developed, its requirements are specified, analyzed, designed, verified and tested.
- ❑ The aim is to produce quality software that satisfies its requirements, that is delivered on time and within the approved budget.
- ❑ **Seminal / Text Book Definition:**
 - ❑ Software engineering is the establishment and use of sound engineering principles in order to obtain economically software that is reliable and works efficiently on real machines.
- ❑ **The IEEE definition:**
 - ❑ Software Engineering is the application of a systematic, disciplined, quantifiable approach to the development, operation, and maintenance of software; that is, the application of engineering to software.

A Layered Technology



Software Engineering

- ❑ Software engineering is a layered technology.
- ❑ The bedrock / fundamental principle that supports software engineering is a **quality focus**.
- ❑ The foundation for software engineering is the **process layer**. Process defines a framework that must be established for effective delivery of software.
 - ❑ The software process forms the basis for management control of software projects and establishes the context in which technical methods are applied, work products (models, documents, data, reports, forms, etc.) are produced, milestones are established, quality is ensured, and change is properly managed.
- ❑ Software engineering **methods** provide the technical how-to's for building software. Methods encompass a broad array of tasks that include communication, requirements analysis, design modeling, program construction, testing, and support.
- ❑ Software engineering **tools** provide automated or semi-automated support for the process and the methods.

A Process Framework

- ❑ A process framework establishes the foundation for a complete software engineering process by identifying a small number of **framework activities that are applicable to all software projects, regardless of their size or complexity.**
- ❑ The process framework is required for representing **common process activities.**

Framework Activities

- Communication
- Planning
- Modeling
 - Analysis of requirements
 - Design
- Construction
 - Code generation
 - Testing
- Deployment

- Each engineering action defined by a framework activity comprises a list of needed
 - work outputs,
 - project milestones,
 - software quality assurance (SQA) points.

Process Framework

Umbrella Activities

Action #1.1

Task sets

Work Task
Milestone
SQA Points

Action #1.k

Task sets

Work Task
Milestone
SQA Points

Action #1.n

Task sets

Work Task
Milestone
SQA Points

- ❑ **Communication:** By communication, customer requirement gathering is done. Communication with consumers and stakeholders to determine the system's objectives and the software's requirements.
- ❑ **Planning:** Establish engineering work plan, describes technical risk, lists resources requirements, work produced and defines work schedule.
- ❑ **Modeling:** Architectural models and design to better understand the problem and for work towards the best solution. The software model is prepared by:
 - ❑ o Analysis of requirements
 - ❑ o Design

- ❑ **Construction:** Creating code, testing the system, fixing bugs, and confirming that all criteria are met. The software design is mapped into a code by:
 - ❑ o Code generation
 - ❑ o Testing
- ❑ **Deployment:** In this activity, a complete or non-complete product or software is represented to the customers to evaluate and give feedback. On the basis of their feedback, we modify the product for the supply of better products.

Umbrella Activities

- In addition, the process framework encompasses a set of **umbrella activities** that are **applicable across the entire software process**.

Umbrella Activities

- Software project tracking and control
- Risk management
- Software quality assurance
- Technical reviews
- Measurement
- Software configuration management
- Reusability management
- Work product preparation and production

THANK YOU

Any questions...?

